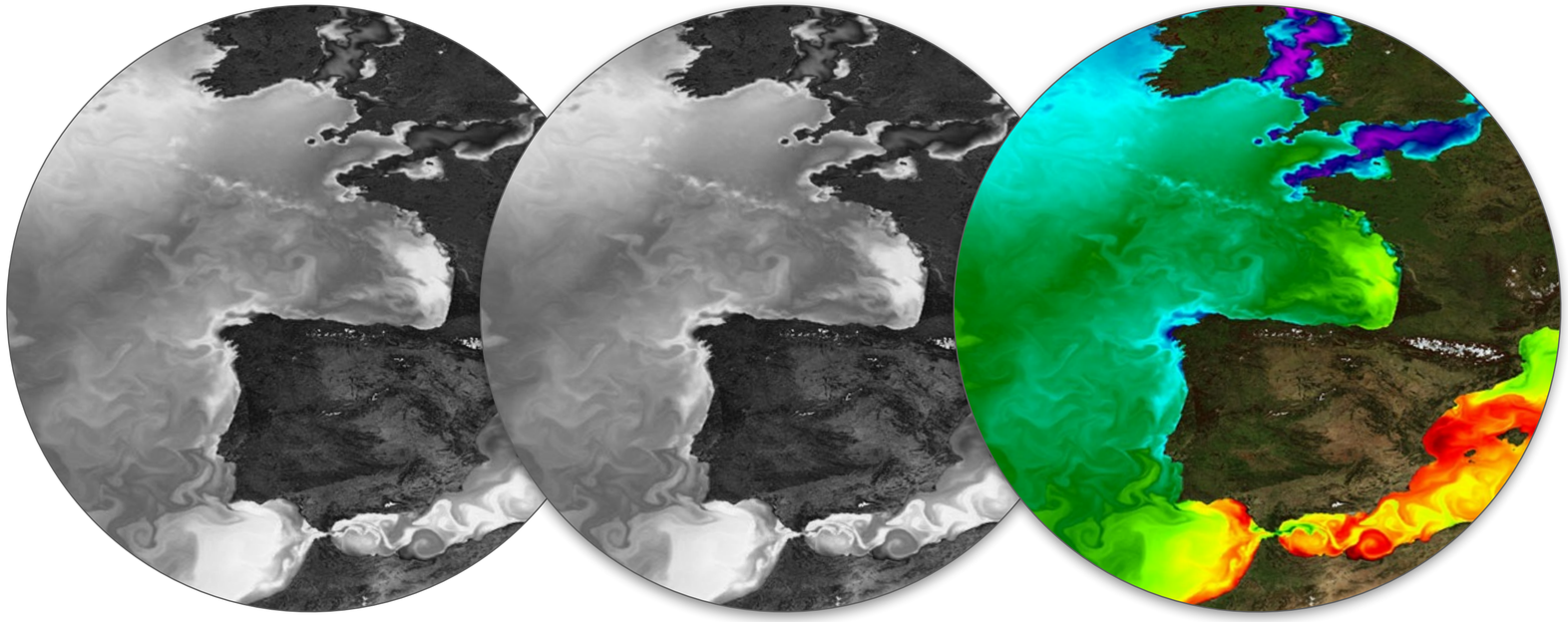
Session 2: Spatial foundations, data acquisition and quality

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Environmental Layers for SDMs



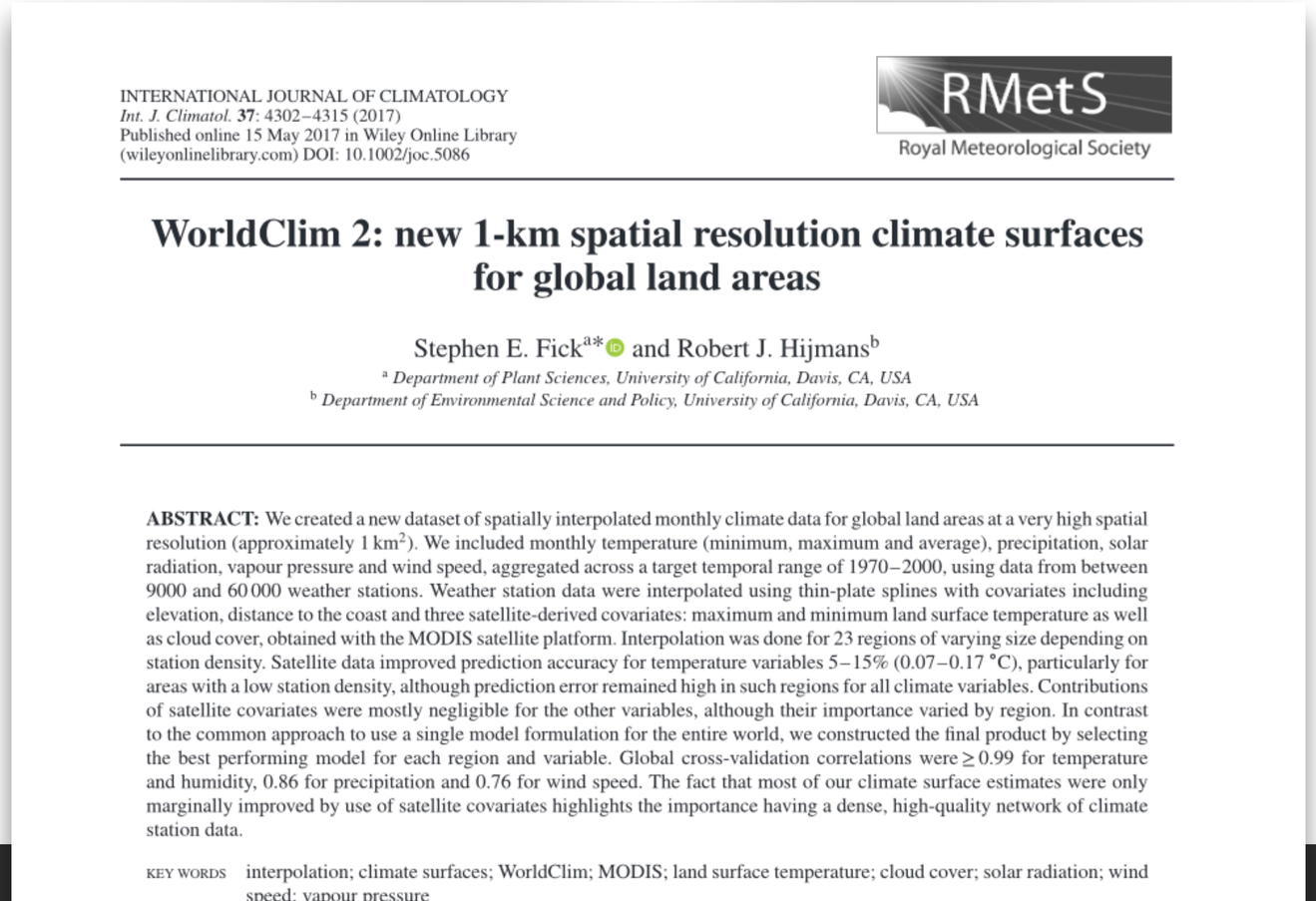
(…) Oxygen Temperature

Environmental data

Allows **linking biodiversity records with the conditions their were exposed to**.

Environmental predictors **typically in the form of digital raster layers**.

For the **terrestrial realm**, GIS databases like **World-Clim** provide **uniform resolution and global coverage**. Incorporates CMIP6 Earth System Models and Shared Socioeconomic Pathways (SSPs) - the standard for terrestrial SDMs.



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RMetS
Royal Meteorological Society

WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas

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ABSTRACT: We created a new dataset of spatially interpolated monthly climate data for global land areas at a very high spatial resolution (approximately 1 km^2). We included monthly temperature (minimum, maximum and average), precipitation, solar radiation, vapour pressure and wind speed, aggregated across a target temporal range of 1970–2000, using data from between 9000 and 60 000 weather stations. Weather station data were interpolated using thin-plate splines with covariates including elevation, distance to the coast and three satellite-derived covariates: maximum and minimum land surface temperature as well as cloud cover, obtained with the MODIS satellite platform. Interpolation was done for 23 regions of varying size depending on station density. Satellite data improved prediction accuracy for temperature variables 5–15% (0.07–0.17 °C), particularly for areas with a low station density, although prediction error remained high in such regions for all climate variables. Contributions of satellite covariates were mostly negligible for the other variables, although their importance varied by region. In contrast to the common approach to use a single model formulation for the entire world, we constructed the final product by selecting the best performing model for each region and variable. Global cross-validation correlations were ≥ 0.99 for temperature and humidity, 0.86 for precipitation and 0.76 for wind speed. The fact that most of our climate surface estimates were only marginally improved by use of satellite covariates highlights the importance having a dense, high-quality network of climate station data.

KEY WORDS interpolation; climate surfaces; WorldClim; MODIS; land surface temperature; cloud cover; solar radiation; wind speed; vapour pressure

For the **marine realm**, **Bio-ORACLE provides uniform resolution**, **global coverage**, and **biologically significant variables**, incorporating also CMIP6 Earth System Models and Shared Socioeconomic Pathways (SSPs).

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DATA ARTICLE

Bio-ORACLE v3.0. Pushing marine data layers to the CMIP6 Earth System Models of climate change research

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Abstract

Motivation: Impacts of climate change on marine biodiversity are often projected with species distribution modelling using standardized data layers representing physi-


This helps evaluate climate models, leads to improvements in the model simulations and provides a better understanding of past, present and future climates.

State-of-the-art

Layers for future conditions need to consider numerous Earth System Models.

ESM are **simulations that resolve global atmospheric and oceanographic conditions** across space and time (e.g., end-of-century SST).

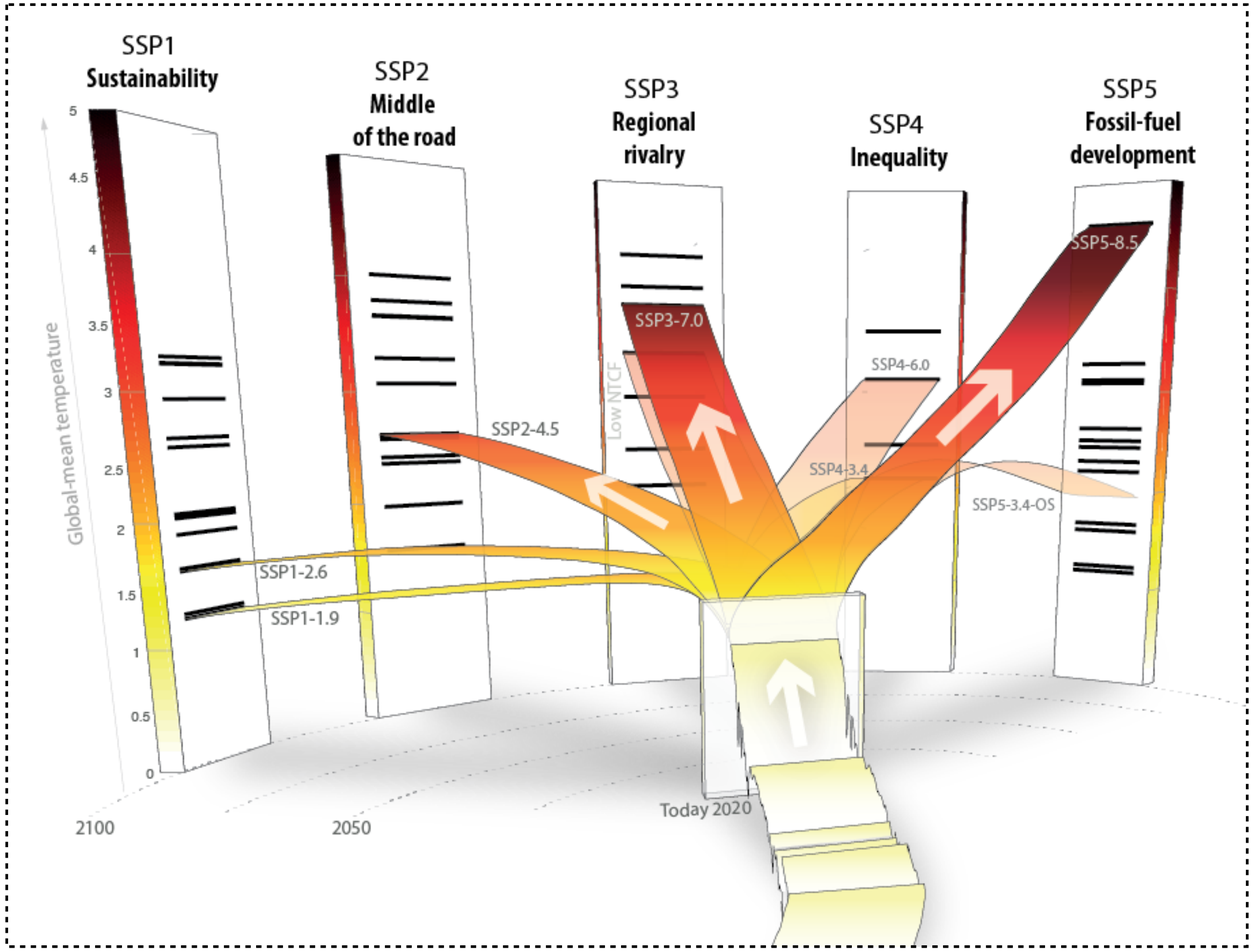
ESM **have inherent uncertainties**.

The **multi-model ensemble** (e.g., averaging different models) can **account for inter-ESM variability**.

State-of-the-art

Layers for future conditions need to **consider data (ESM) from the 6th phase of the Coupled Model Inter-comparison Project.**

CMIP6 (1) is **more accurate in capturing the spatial and temporal variability of oceanographic conditions** and (2) **outputs the SSP scenarios**, which are the new narratives of global change.

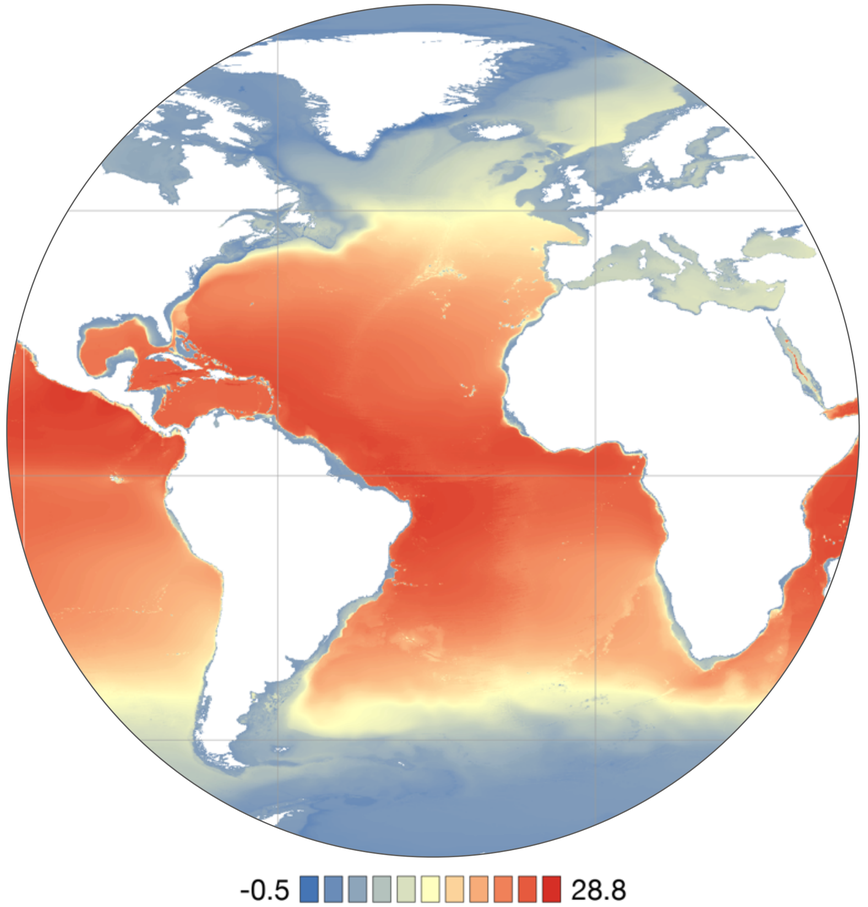


The new **SSP use up-to-date socioeconomic assumptions** and **encompass the Paris Agreement expectations**, allowing the development of **projections aligned with international climate policies**.

State-of-the-art

Moving beyond just temperature: While temperature is a key driver, relying on it exclusively can truncate species' niche. Biologically meaningful models must incorporate variables like oxygen, primary productivity, and pH.

Failing to consider these variables in climate change projections might **misinterpret the exposure of biodiversity to future detrimental conditions**.



Difference between surface and bottom temperature

The 3D Ocean: Unlike terrestrial models, marine species are exposed to depth gradients. Sea bottom layers are key for **benthic species** (e.g., corals or fish).

		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		1 (1970-2020)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)	4 (2020-2100)
1	**Air temperature**							
2	**Precipitation**							

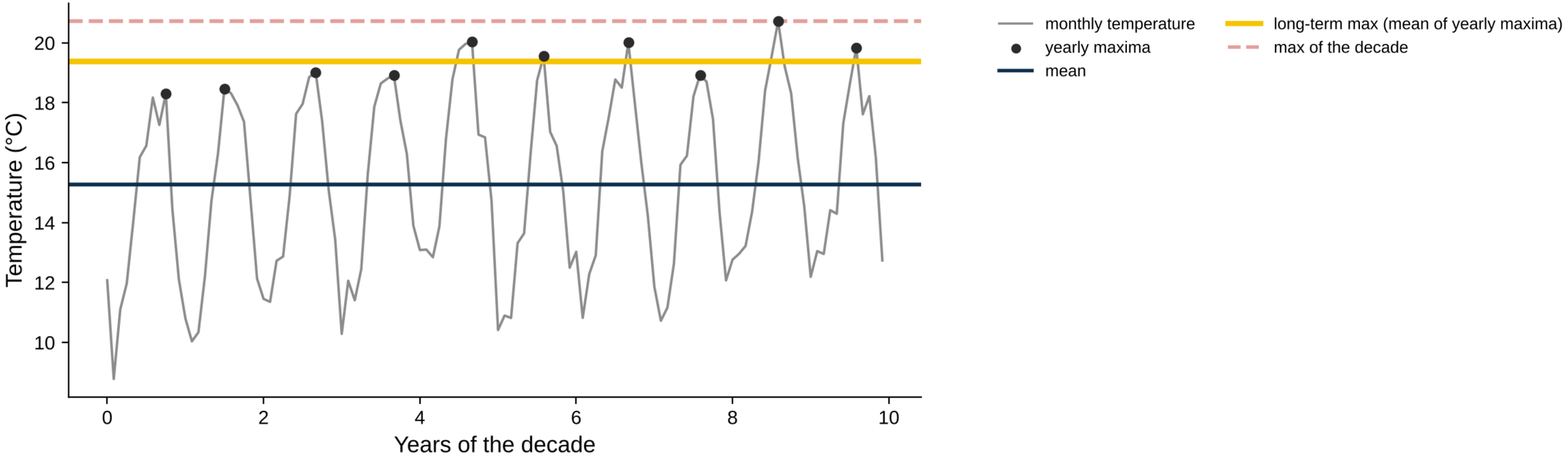
WorldClim 2.1: 2 variables from 1970-2100, by 20-year averages at ~1km **resolution**, **covering the span of SSPs**, by **ensembling 23 ESMs of CMIP6.**

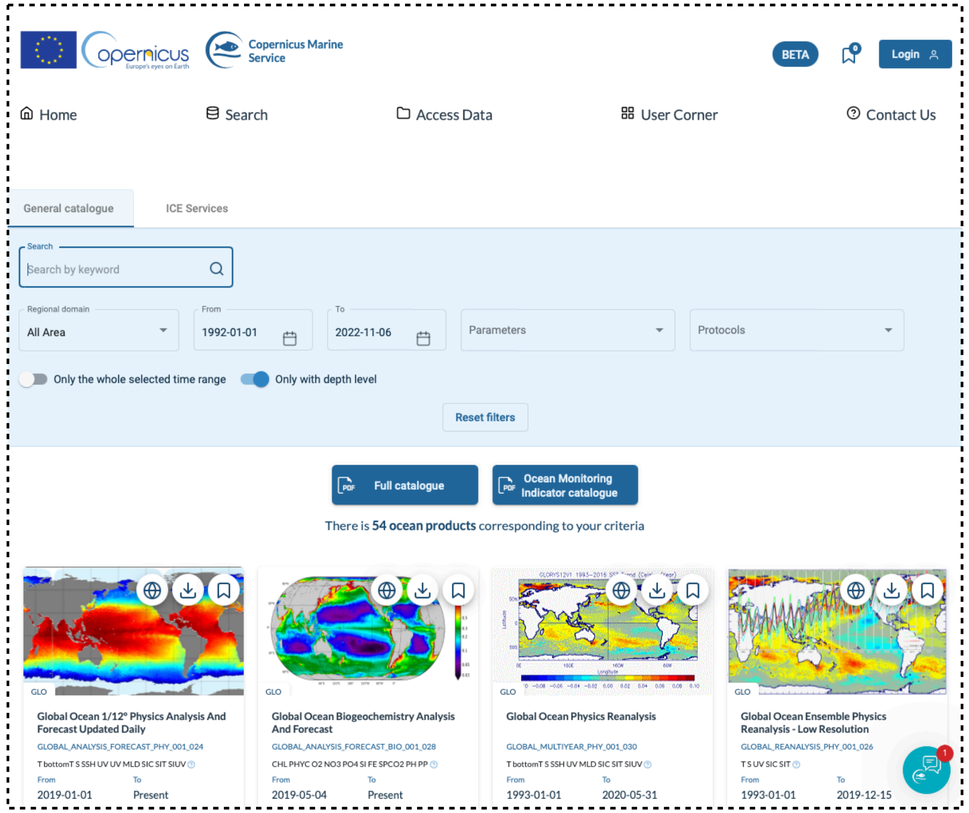
		Present	SSP1-1.9	SSP1-2.9	SSP2-4.5	SSP3-7.0	SSP4-6.0	SSP5-8.5
		2 (2000-2020)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)	8 (2020-2100)
1	**Temperature**	■	■	■	■	■	■	■
2	**Salinity**	■	■	■	■	■	■	■
3	**Sea Ice Cover**	■	■	■	■	■	■	■
4	**Sea Ice Thickness**	■	■	■	■	■	■	■
5	**Sea Water Velocity**	■	■	■	■	■	■	■
6	**Mixed Layer Depth**	■	■	■	■	■	■	■
7	**Diffuse Attenuation Coefficient**	■						
8	**PAR**	■						
9	**PAR at Bottom**	■						
10	**Oxygen**	■	■	■	■	■	■	■
11	**pH**	■	■	■	■	■	■	■
12	**Iron**	■	■	■	■	■	■	■
13	**Phosphate**	■	■	■	■	■	■	■
14	**Nitrate**	■	■	■	■	■	■	■
15	**Silicate**	■	■	■	■	■	■	■
16	**Primary Production**	■	■	■	■	■	■	■
17	**Phytoplankton**	■	■	■	■	■	■	■
18	**Chlorophyll**	■	■	■	■	■	■	■

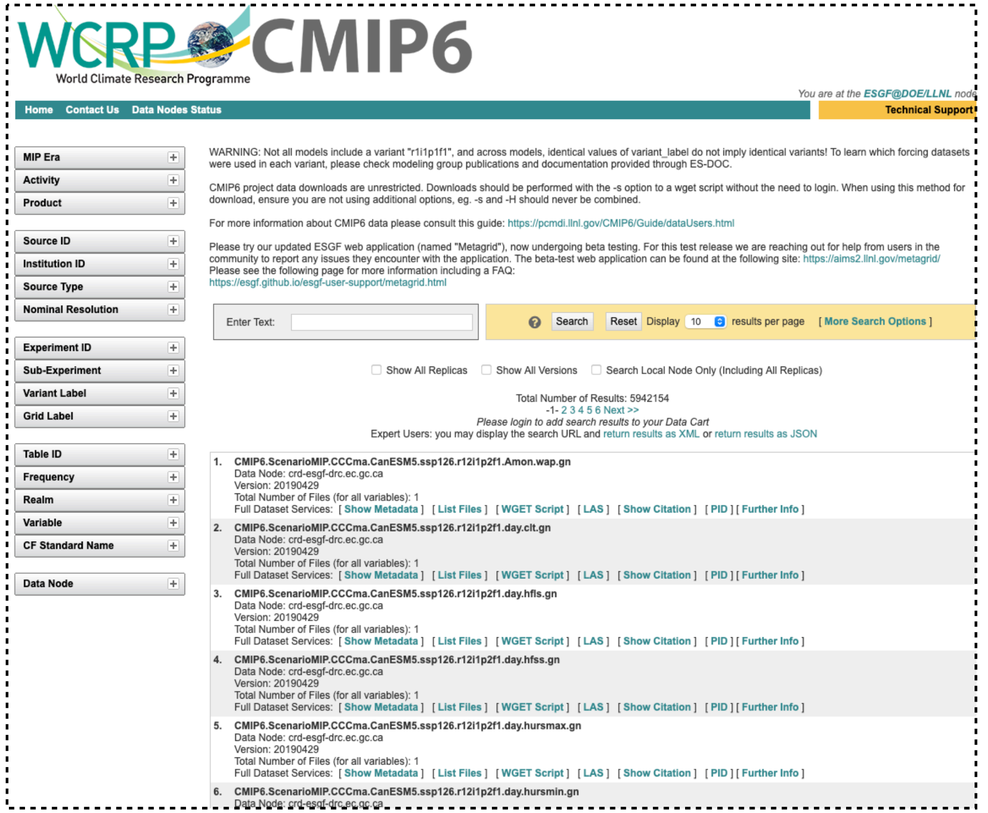
Bio-ORACLE: **18 essential variables** for the surface and along the seafloor, **from 2000 to 2100**, **per decade**, at ~5km **resolution**, **covering the span of SSPs**, by **ensembling 10 ESMs of CMIP6**.

Six statistics per variable

Per decade and variable: the mean; the maximum and minimum of the decade; the long-term maximum and minimum (the average of the yearly extremes); and the range.

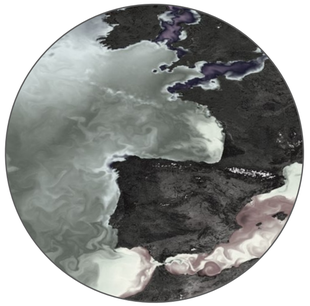






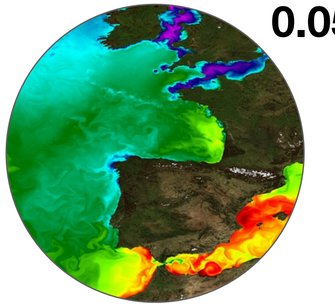
Examples of Data Sources:

Copernicus Marine Service and World Climate Research Programme



Raw data
1993-2025

Subsetting data and
resampling to desired resolution



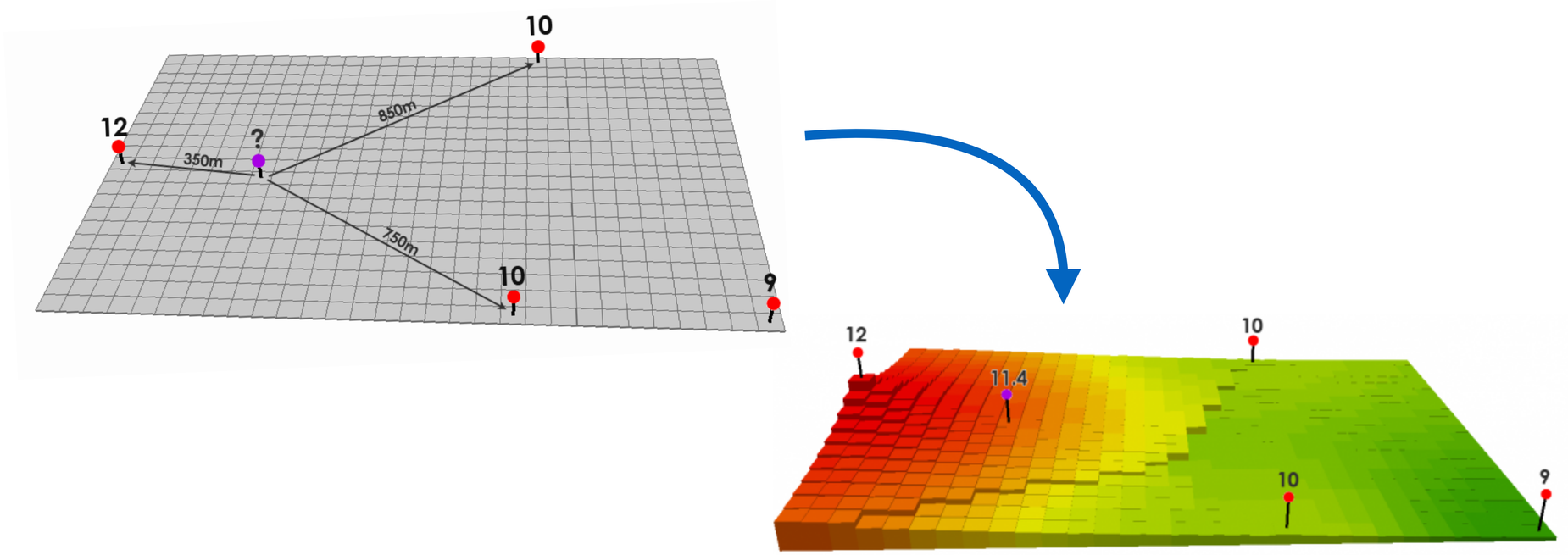
0.05°

Predictor Layer
2010-2020

For present-day conditions, layers are developed by subsetting raw data to the **desired decade** and **resampling** (interpolating) to the desired resolution.

e.g.,

layer 1: Global maximum sea surface temperature of 2010-2020



Interpolation

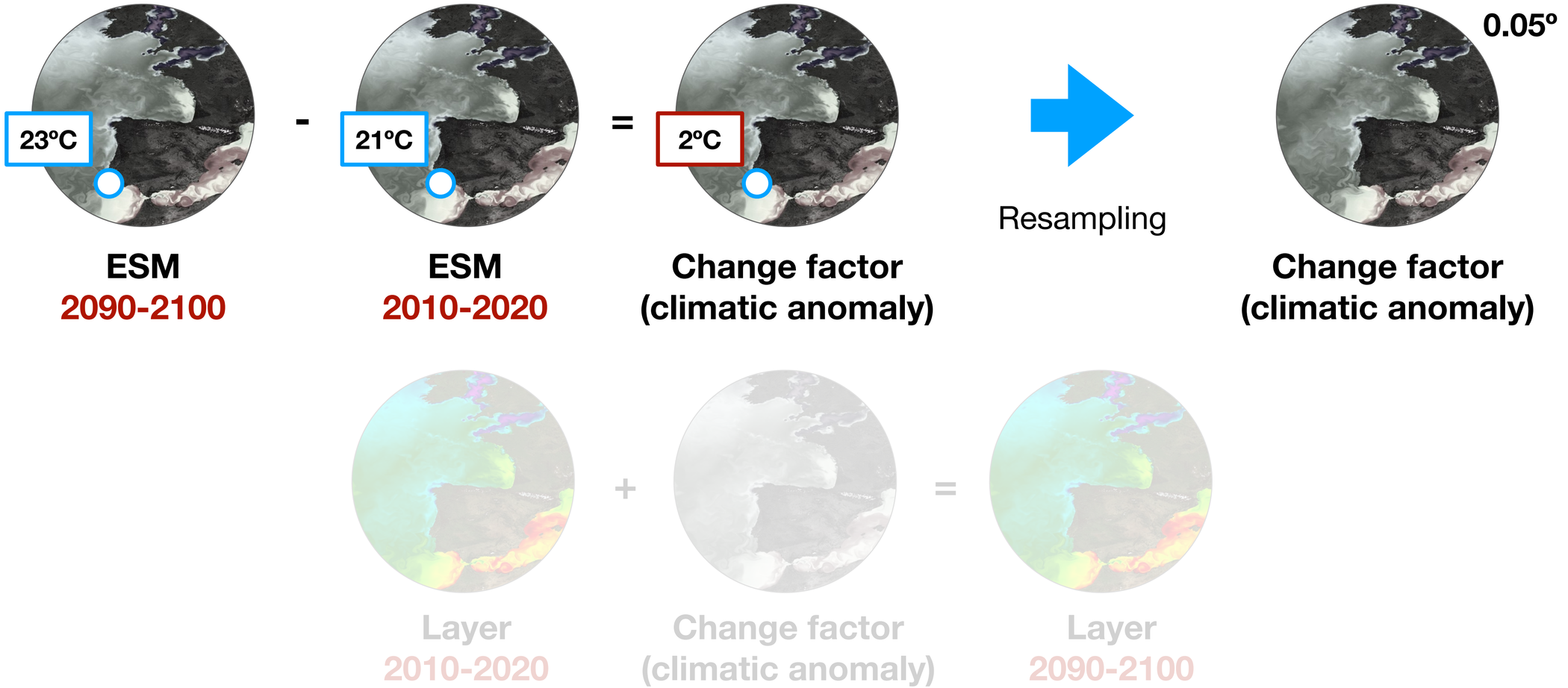
Raw **data are collected** from numerous sources (e.g., buoys, weather stations) and their information are spatially interpolated to infer the distribution of the environmental gradients for the whole region - **estimates unknown points**.

e.g., Inverse Distance Weighting algorithm.

For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

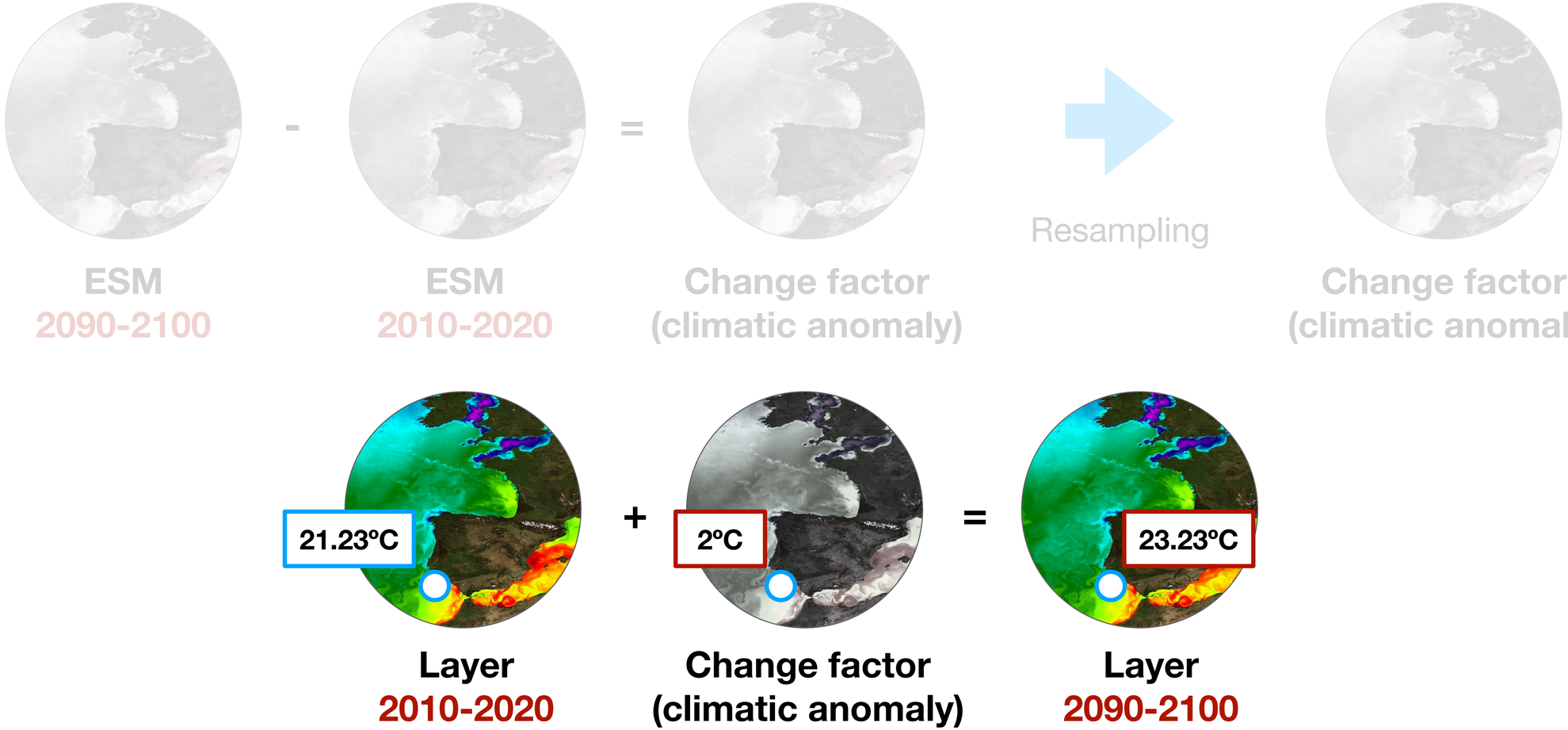
layer 2: Global maximum sea surface temperature of 2090-2100



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100



For the future, use of the **change-factor technique**, applying the magnitude of climate change (projected by ESMs) to the present-day layers.

e.g.,

layer 2: Global maximum sea surface temperature of 2090-2100

2 min. PITCH

Papers: P19, P20

10 minutes

Read the abstract and the introduction and interpret the main figures.

Question. What problem did the authors tackle? (one sentence)

Approach. What did they do, and with which data? (one sentence)

Key finding. The one result to remember, ideally with the paper's main figure.

Why it matters today?